

Industrial Internship Report on "Iris Flower Classification Using Machine Learning"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Iris Flower Classification Using Machine Learning." This project uses Machine Learning to classify Iris flowers into Setosa, Versicolor, and Virginica species based on flower measurements. The model was developed using Python and Random Forest Classifier to achieve accurate predictions.

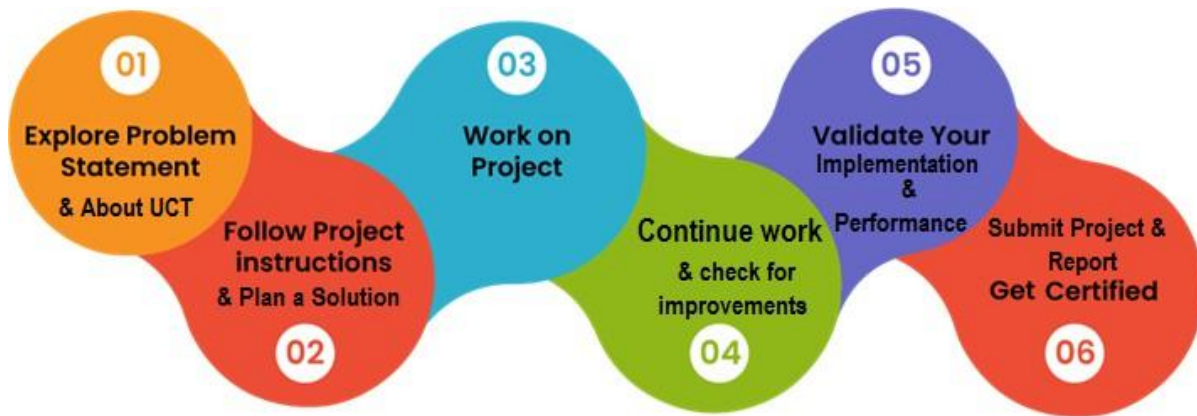
This internship gave me a very good opportunity to get exposure to Industrial problems.

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1 Preface

This report presents the work completed during my Industrial Internship offered by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship provided me with an excellent opportunity to gain practical exposure to industry-oriented projects and enhance my technical skills.



I would like to express my sincere gratitude to ****upskill Campus****, ****The IoT Academy****, and ****UniConverge Technologies Pvt Ltd (UCT)**** for providing me with this valuable internship opportunity. I am thankful to all the mentors and coordinators who guided me throughout the internship program.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT)**, **Cyber Security**, **Cloud computing (AWS, Azure)**, **Machine Learning**, **Communication Technologies (4G/5G/LoRaWAN)**, **Java Full Stack**, **Python**, **Front end** etc.



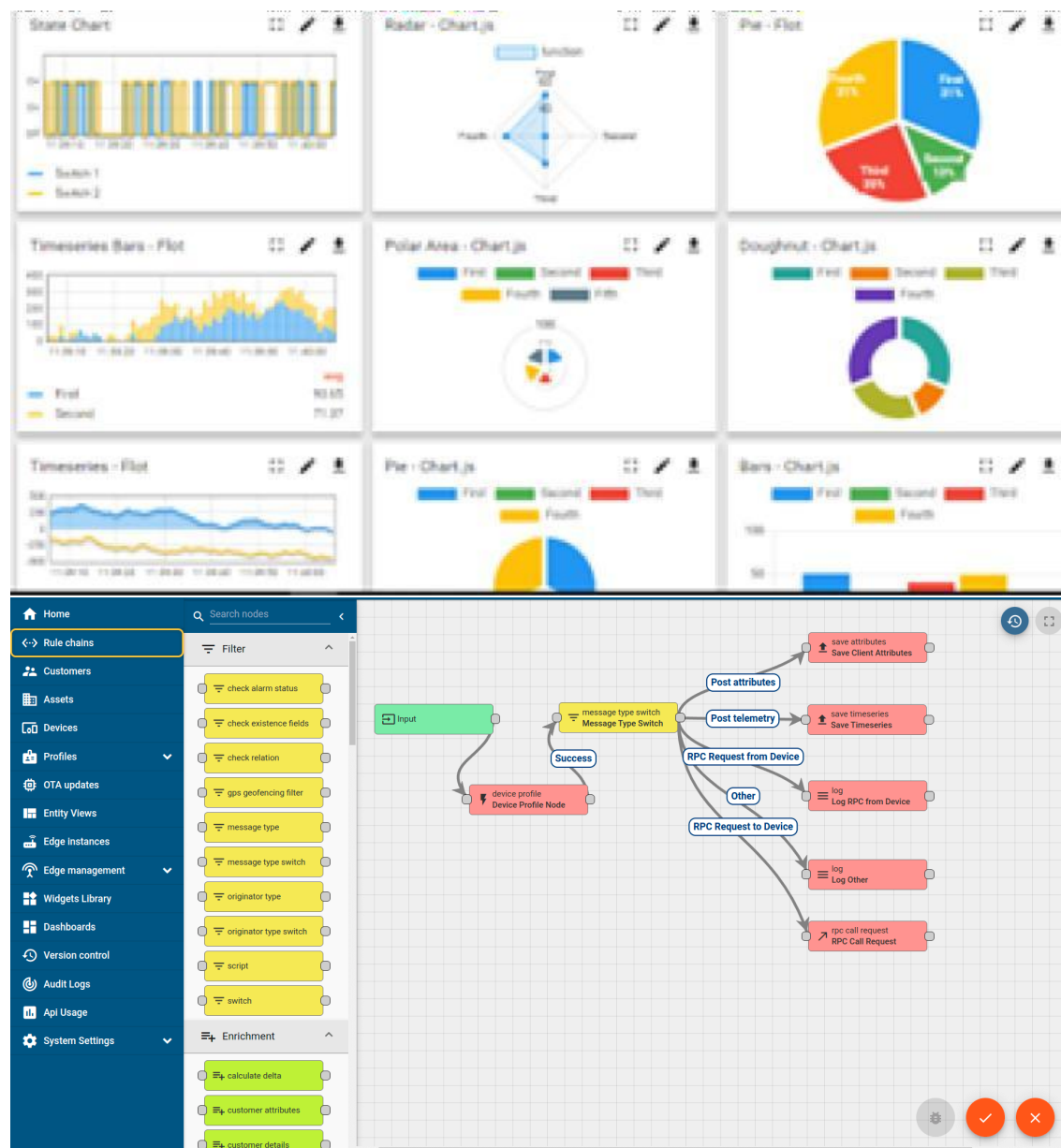
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i





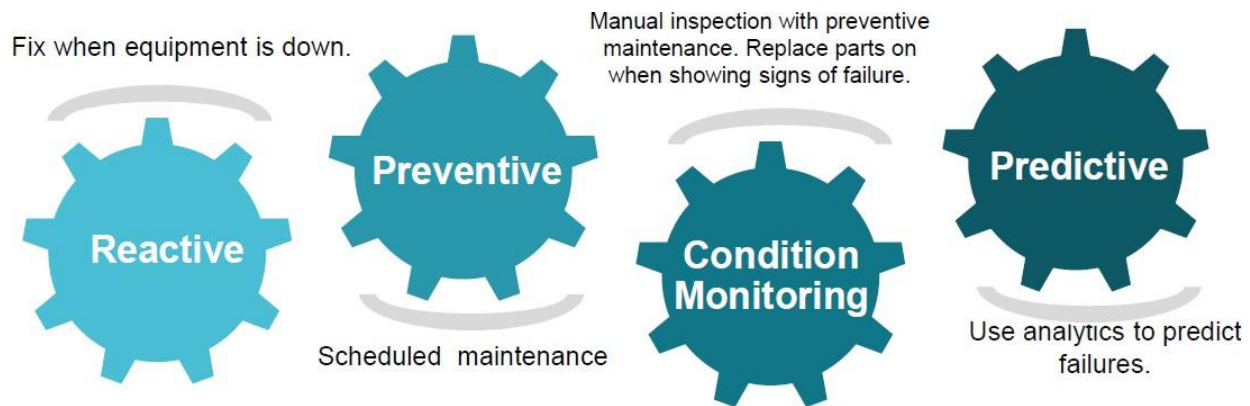
iii.

based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

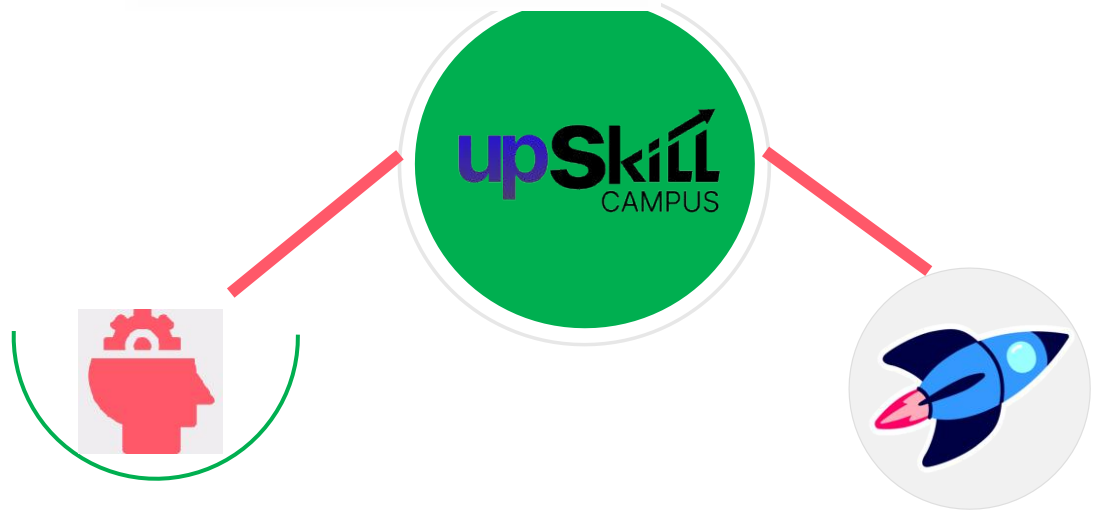
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

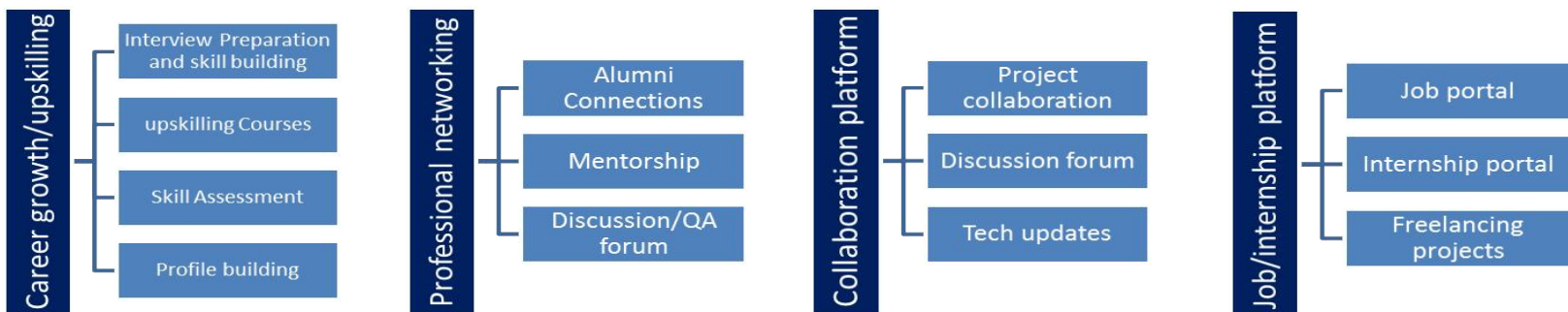
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Python Software Foundation, Python Documentation, <https://docs.python.org/>
- [2] Scikit-Learn Documentation, Machine Learning in Python, <https://scikit-learn.org/>
- [3] Fisher, R.A., Iris Flower Dataset, UCI Machine Learning Repository, <https://archive.ics.uci.edu/>

2.6 Glossary

Terms	Acronym
Artificial Learning	AI
Machine Learning	ML
Dataset	DS
Python	PY
Scikit-Learn	SKL

3 Problem Statement

In the assigned problem statement, the objective was to develop a Machine Learning model capable of classifying Iris flowers into three species namely Setosa, Versicolor, and Virginica. The classification was based on flower measurements such as sepal length, sepal width, petal length, and petal width. The main challenge was to build an accurate prediction model using Machine Learning techniques and evaluate its performance on test data. This project aimed to demonstrate how Machine Learning can be used to solve real-world classification problems efficiently and accurately.

4 Existing and Proposed solution

Existing flower classification methods mainly rely on manual observation and traditional statistical techniques. These methods require human expertise, are time-consuming, and may lead to classification errors when dealing with large datasets. Some traditional approaches also provide lower accuracy and limited scalability.

Limitations of Existing Solutions:

Requires manual analysis and expert knowledge.

Time-consuming for large datasets.

Possibility of human errors.

Lower efficiency and scalability.

4.2 Report Submission (Github Link):

<https://github.com/prabavathi616/upskillCampus>

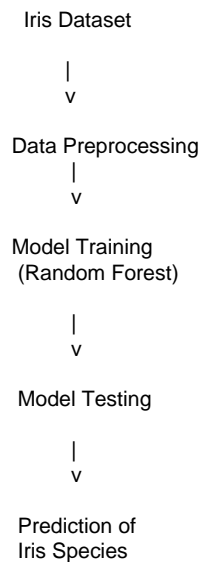
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5 Proposed Design/ Model

The developed Machine Learning model successfully classified Iris flowers into Setosa, Versicolor, and Virginica species using flower measurements. The Random Forest Classifier produced accurate predictions and demonstrated good performance on the test dataset. The project achieved its objective of automating the classification process and reducing manual effort. The outcome of this project enhanced my understanding of Machine Learning algorithms, data preprocessing, model training, and performance evaluation.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM



6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

During this internship, I gained practical knowledge of Machine Learning concepts and their real-world applications. I learned how to work with datasets, perform data preprocessing, train Machine Learning models, and evaluate their performance using Python and Scikit-Learn. I also improved my problem-solving, analytical thinking, and technical documentation skills while completing the project and preparing the report. This internship helped me understand the complete project development lifecycle and increased my confidence in working on Data Science and Machine Learning projects. The knowledge and experience gained during this internship will support my academic growth and help me build a successful career in the field of Artificial Intelligence and Machine Learning.

8 Future work scope

The current project successfully classifies Iris flowers using a Machine Learning model. In the future, the project can be enhanced by using larger and more diverse datasets to improve model performance. Advanced Machine Learning and Deep Learning algorithms can be implemented to achieve higher accuracy. A user-friendly web application can also be developed to allow users to upload flower measurements and receive predictions instantly. Additional flower species can be included in the dataset, making the system more practical and scalable for real-world applications. Due to time limitations, these enhancements were not implemented during the internship but can be considered for future development.